# Computer Science

*Foundations, Research & Discovery*

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# What Is Computer Science?

**The scientific and practical study of computation**

**CS asks:**

“What can be computed?”
“How efficiently?”
“How reliably?”
“How do we apply it?”

### Artificial Intelligence

Making machines think & learn

### Software Engineering

Building reliable systems

### Networking & Security

Connecting & protecting data

### Theory & Algorithms

Mathematical foundations of computing

# The Landscape of Computer Science

*A discipline spanning pure mathematics to everyday applications*

### Theory of Computation

Automata, complexity, computability, logic

### Artificial Intelligence

ML, deep learning, NLP, computer vision

### Data Science & Databases

Big data, analytics, knowledge discovery

### Systems & Architecture

OS, compilers, hardware, distributed systems

### Networks & Cybersecurity

Internet protocols, cryptography, privacy

### Computational Mathematics

Graph theory, geometry, combinatorics, numerics

PART TWO

# Research in Computer Science

*What does it mean to do CS research — and why does it matter?*

# What Does CS Research Look Like?

### Theoretical

- Proofs & formal models
- Complexity & computability
- Graph theory & combinatorics
- No experiments needed — math is the proof

### Experimental

- Build systems & benchmark
- Empirical performance tests
- Dataset experiments
- Statistical evaluation of results

### Applied / Interdisciplinary

- Solve real-world problems
- Cross-domain (medicine, policy…)
- Industry collaboration
- Turn theory into tools & products

# From Question to Contribution

*The typical arc of a full university-level CS research project*

**01**

**Problem**

Identify an open question or bottleneck

**02**

**Literature**

Survey existing work — what's known?

**03**

**Design**

Invent an algorithm, model or proof strategy

**04**

**Evaluate**

Prove correctness or run experiments

**05**

**Publish**

Paper at a conference or journal

PART THREE

# Dr. Mudassir Shabbir

*Research Areas at LUMS*

Computational Geometry · Graph Theory · Network Control · Graph ML · Applied Data Science

# Dr. Mudassir Shabbir

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PhD in Computer Science
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*Rutgers Honors Fellow 2011–12*

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## Research Fingerprint

- **Algorithmic & Discrete Geometry**
- **Combinatorics & Extremal Graph Theory**
- **Network Controllability · Zero Forcing**
- **Graph Machine Learning & GNNs**
- **Applied Data Science & Smart Policing**

# Research Area 1 — Algorithmic & Discrete Geometry

## Core Idea

**How do you efficiently represent, query, and reason about geometric data?**

Dr. Shabbir developed new methods for computing succinct (compact) representations of large datasets — finding the smallest structure that still captures essential statistical properties.

**Key concept:** Tukey depth & ray-shooting depth — measuring how 'central' a point is within a cloud of data points.

## Applications

- Nonparametric statistics
- Outlier detection in high-dim data
- Computing hitting sets for convex ranges

## Key Work

- "Ray-Shooting Depth in $R^2$ — Algorithms & Applications" (ESA 2011)
- "Computing Small Hitting Sets for Convex Ranges" (with Langerman & Steiger)

# Research Area 2 — Combinatorics & Graph Theory

## What is Graph Theory?

A graph is a set of nodes (vertices) connected by edges — the universal language for modelling relationships.

**Combinatorics** asks: how many? under what constraints?

**Extremal Graph Theory** seeks the maximum or minimum number of edges a graph can have while still satisfying a given property.

### Spectral Graph Theory

Studying graphs via eigenvalues of their matrices — graphs 'determined by their spectra'

### Graph Decontamination

Cleaning or traversing a contaminated graph with minimum agent count

### Extremal Problems

When must a graph of a given density contain a specific substructure?

### Skew Adjacency Spectra

Directed-graph generalisation: oriented graphs sharing generalised skew spectra

# Research Area 3 — Network Controllability & Zero Forcing

## The Core Question:

*"Can we steer every node in a network to any desired state by controlling just a small subset of nodes (leaders)?"*

## Zero Forcing — a combinatorial tool

Colour some vertices blue. Rule: if a blue vertex has exactly ONE white neighbour, colour it blue too.
If blue spreads to ALL vertices → the initial set is a **Zero Forcing Set (ZFS)**
Minimum ZFS size ≈ minimum number of leader nodes needed for strong structural controllability.

### Controllability Backbone

Identify the minimal set of edges that preserve full network controllability — useful for resilient network design.

### Distance-Based Bounds

New bounds on Strong Structural Controllability using distances between nodes (IEEE TAC 2023).

### Edge Augmentation

Add edges to densify a network while guaranteeing controllability doesn't degrade.

# Research Area 4 — Graph Machine Learning & GNNs

## Graph Neural Networks (GNNs)

Traditional ML treats data as flat tables. But the world is **graph-structured** — social networks, molecules, brains, roads.

GNNs learn representations by aggregating information along edges.

**Dr. Shabbir's angle:** use network control theory (zero forcing, average controllability) to build BETTER node features for GNNs.

### NeuroGraph

Benchmarks for graph ML in brain connectomics — modelling brain regions as graph nodes (NeurIPS 2023)

### Ego-Centric Spectral Augmentation

Enhanced GNNs using subgraph spectral embeddings to capture local graph structure

### Graph Unlearning

Survey: How to 'forget' training data from a trained GNN — critical for privacy-preserving AI

### Control-Theoretic Features

Average controllability & rank encoding as graph features to improve social-network classification

# Research Area 5 — Applied Data Science & Real-World Impact

### Smart Policing & Crime Analytics

- Optimising police substation locations using graph-based road distances
- Detecting spatial crime patterns in Lahore (Smart City research)
- Data-driven law enforcement resource allocation

### Natural Language Processing

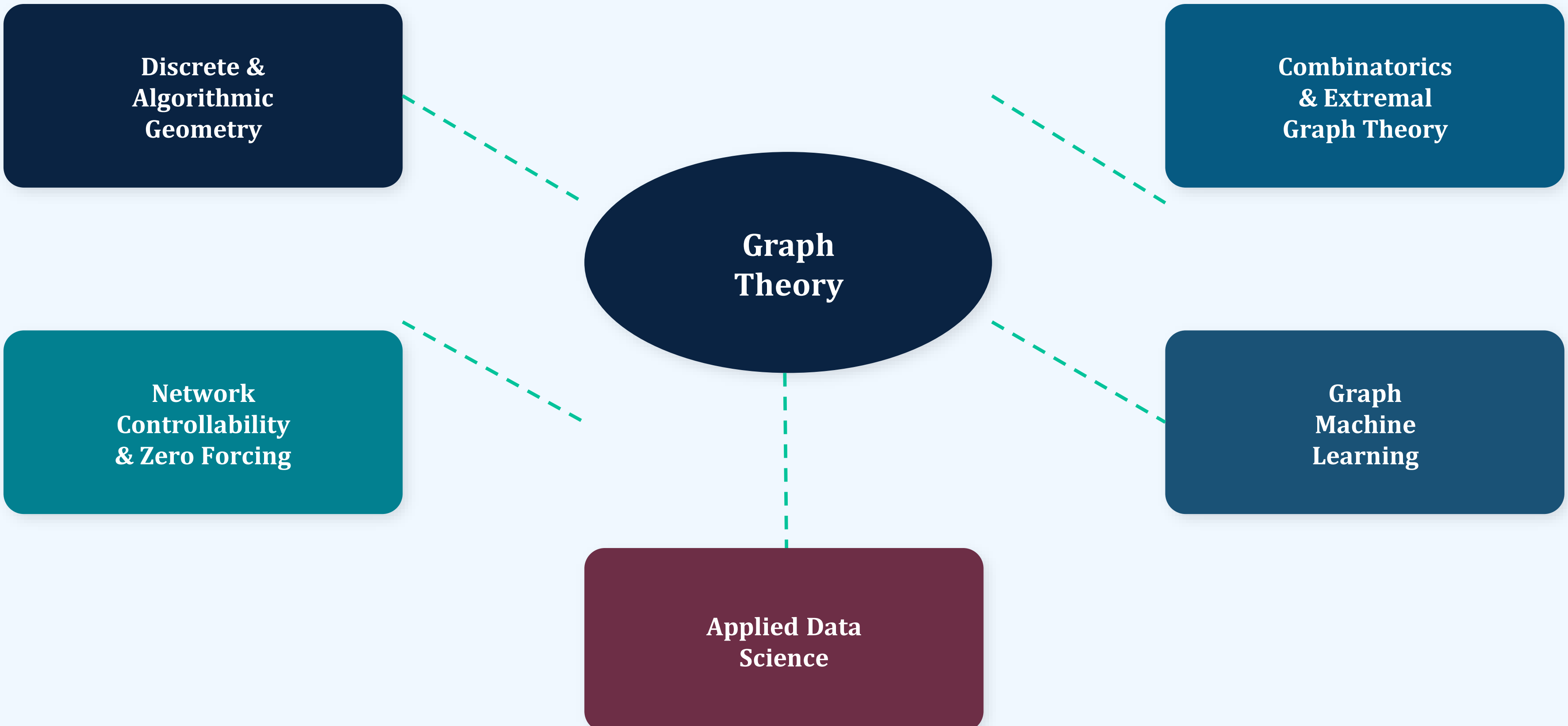
- Conversational AI: automatic dialogue generation & evaluation
- Misinformation detection on social media (Pakistan study)
- Speech: fake-audio detection in resource-constrained settings

### Scalable Graph Descriptors

- MSDGSD: processing very large graphs efficiently for ML
- Graph representation for edge-stream data
- Bridging distributed systems with graph learning

# The Big Picture: How It All Connects

*Dr. Shabbir's research forms one coherent web — from pure math to real-world impact*



# Key Takeaways

1. CS is a vast field spanning pure mathematics to real-world engineering
2. Research means asking precise questions and finding verifiable answers
3. Dr. Mudassir works at the exciting intersection of discrete math, control theory, and modern AI
4. Zero forcing is a single elegant idea that unlocks insights in all three domains